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Metabolic Response To Injury

BASIC CONCEPTS IN HOMEOSTASIS:

- Homeostasis is the foundation of normal physiology
- 'Stress-free' perioperative care helps to preserve homeostasis following elective surgery.
- Resuscitation, surgical intervention & critical care can return the severely injured patient to a situation in which homeostasis becomes possible once again.

In elective surgery:

- 1- Minimal access surgery.
- 2- 'Stress free' perioperative care.

In emergency surgery:

- 1- Resuscitation (artificially homeostatic responses).
- 2- Critical care: (provide organ support).

The response to injury is graded:

The more severe the injury, the greater the response in metabolic & immunological changes:

Following elective surgery of intermediate severity, there may be a transient & modest rise in:

- Temp.**
- HR.**
- RR.**
- Energy expenditure.**
- WBC count.**

Following major trauma/sepsis, these changes are accentuated, resulting in:

- **Systemic Inflammatory Response Syndrome (SIRS)**
- **Hypermetabolism,**
- **Marked catabolism,**
- **Shock**
- **Multiple Organ Dysfunction (MODS).**

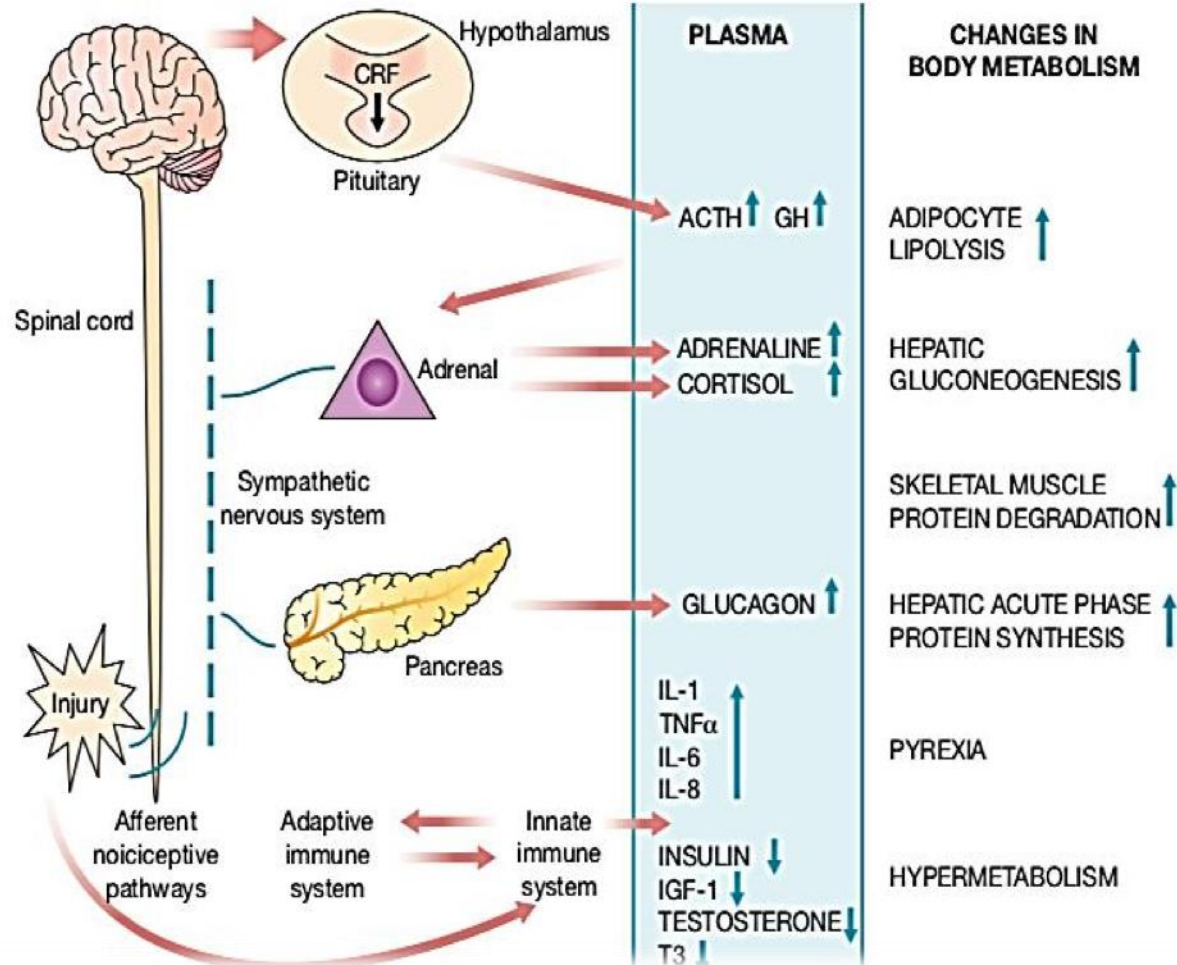
The immunological sequelae of major injury:

evolve from a **proinflammatory** state driven primarily by the innate immune system:

(macrophages, neutrophils, dendritic cells) into a **compensatory anti inflammatory response syndrome (CARS)** characterised by:

- Suppressed immunity.
- & Diminished resistance to infection.

MEDIATORS OF THE METABOLIC RESPONSE TO INJURY



The classical neuroendocrine pathways of the stress response consist of afferent nociceptive neurones, the spinal cord, thalamus, hypothalamus & pituitary

Corticotrophin releasing factor (CRF)⁹ released from the *hypothalamus* increases adrenocorticotrophic hormone (ACTH) release from the anterior *pituitary*.

ACTH then acts on the *adrenals* to increase the secretion of cortisol.

Hypothalamic activation of the sympathetic nervous system causes release of adrenaline & also stimulates release of glucagon.

There are, many other players:

- 1- Alterations in insulin release & sensitivity.
- 2- Hypersecretion of prolactin & GH.
- 3- Inactivation of peripheral thyroid hormones & gonadal function.

Functions of GH :

- 1- Lipolytic.
- 2- Insulin antagonising.
- 3- Proinflammatory properties.

Neuroendocrine response to injury:

●● Acute Phase: (Changes for short- term survival)
characterised by: an actively secreting pituitary & elevated counter-regulatory hormones (cortisol, glucagon, adrenaline).

●● Chronic Phase: (Changes contribute to chronic wasting):

- Hypothalamic suppression.
- Low serum levels of the respective target organ hormones.

The ***Innate Immune System*** (principally macrophages)
interacts in a complex manner with:
The ***Adaptive Immune System*** (T cells, B cells)
in cogenerating the metabolic response to injury.

Proinflammatory cytokines: including

- 1- Interleukin1 (IL1).
- 2- Tumour necrosis factor alpha (TNFa).
- 3- IL6.
- 4- IL8.

are produced within the first 24 hours and act:

- 1- On the hypothalamus to cause pyrexia.
- 2- Augment the hypothalamic stress response.
- 3- Act on skeletal muscle to induce proteolysis.
- 4- Inducing acute phase protein production in the liver.
- 5- Development of peripheral insulin resistance.

Systemic inflammatory response syndrome

- Is driven initially by proinflammatory cytokines (e.g. IL-1, IL-6 & TNF α)
- Is followed rapidly by increased plasma levels of cytokine antagonists & soluble receptors (e.g. IL-1Ra(interleukin1 receptor antagonist), TNF-sR)
- If prolonged or excessive may evolve into a counter inflammatory response syndrome

THE METABOLIC STRESS RESPONSE TO SURGERY & TRAUMA:

THE 'EBB & FLOW' MODEL

THE EBB PHASE

(at the time of injury & lasts for 24–48 hours.)

It may be attenuated by proper resuscitation, but not completely abolished.

The ebb phase is characterised by:¹⁷

- 1- Hypovolaemia.
- 2- Decreased basal metabolic rate
- 3- Reduced cardiac output.
- 4- Hypothermia.
- 5- Lactic acidosis.

Hormones regulating the ebb phase are:

- 1- Catecholamines.
- 2- Cortisol.
- 3- Aldosterone (following activation of the renin–angiotensin system).

The magnitude of this neuroendocrine response depends on the:

- 1- Degree of blood loss.**
- 2- Stimulation of somatic afferent nerves at the site of injury.**

The main physiological role of the ebb phase is to conserve both:

- 1- Circulating volume.**
- 2- Energy stores.**

for recovery & repair.

THE FLOW PHASE

Following resuscitation, the ebb phase evolves into a hypermetabolic flow phase, which corresponds to SIRS.

This phase involves the mobilisation of body energy stores for recovery & repair, & the subsequent replacement of lost or damaged tissue.

It is characterised by:

- 1- Tissue oedema (from vasodilatation & increased capillary leakage).
- 2- Increased basal metabolic rate (hypermetabolism).
- 3- Increased cardiac output.
- 4- Raised body temperature
- 5- Leukocytosis.
- 6- Increased oxygen consumption.
- 7- Increased gluconeogenesis.

The flow phase may be subdivided into:

- Catabolic phase: (3–10 days.)
- Anabolic phase: (for weeks) recovery & repair.

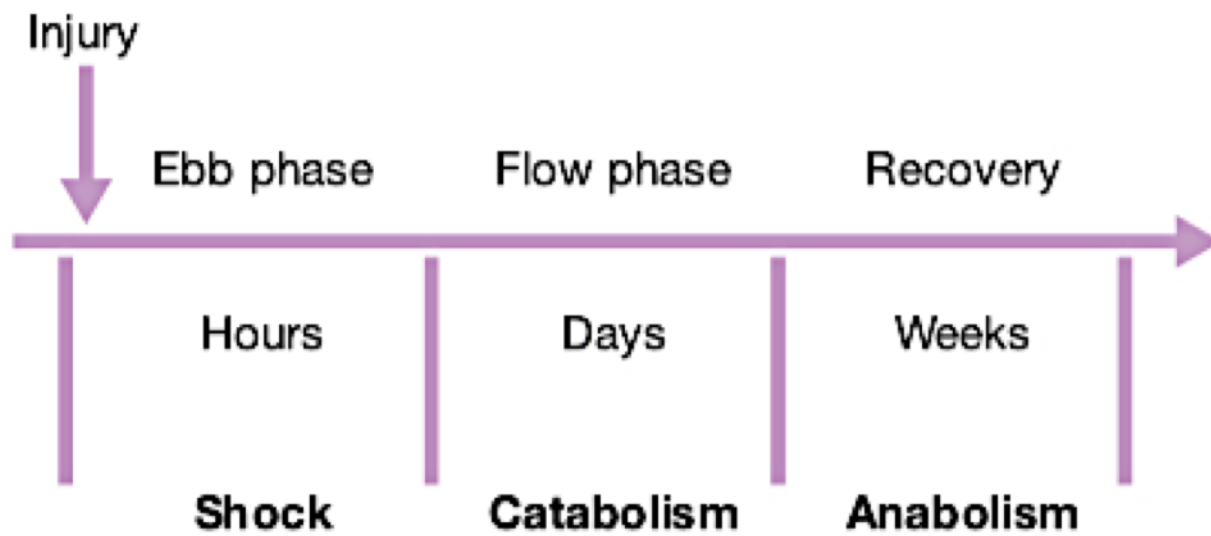
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During the catabolic phase, the increased production of counterregulatory hormones (including catecholamines, cortisol, insulin and glucagon) and inflammatory cytokines (e.g. IL1, IL6 and TNFa) results in:

- * fat & protein mobilisation, leading to weight loss & increased urinary nitrogen excretion.

- * The increased production of insulin at this time is associated with significant insulin resistance and, therefore, poor glycaemic control.

Obviously, the development of complications will further aggravate:

the *neuroendocrine & inflammatory stress responses*, thus creating a *vicious catabolic cycle*.



THANK YOU

